

Selecting an Analog Devices RF Low Noise Amplifier

by Eamon Nash

INTRODUCTION

Figure 1 shows the frequency response of HMC8413, a 10MHz to 9GHz low noise amplifier (LNA). In addition to gain and noise figure, compression, second order and third order distortion are included. The device exhibits a flat response across more than six octaves. While the LNA noise figure and frequency range are critical, all of the other specifications shown in Figure 1 can impact the performance of the LNA in its end application. This can make the process of selecting an RF LNA challenging. In addition, it can be difficult to differentiate all of the devices in the LNA portfolio of Analog Devices Inc. with more than 200 generics.

This application note is intended to help RF circuit designers understand and select devices from Analog Devices large and diverse portfolio of RF LNAs. It begins with an overview of the Analog Devices portfolio. This includes a discussion of the relative vintage of different product families and guidelines on part number nomenclature. From there, various considerations that affect the choice of LNA are discussed. The document is structured so that it can be read from start to finish or can be used as a reference source on specific topics.

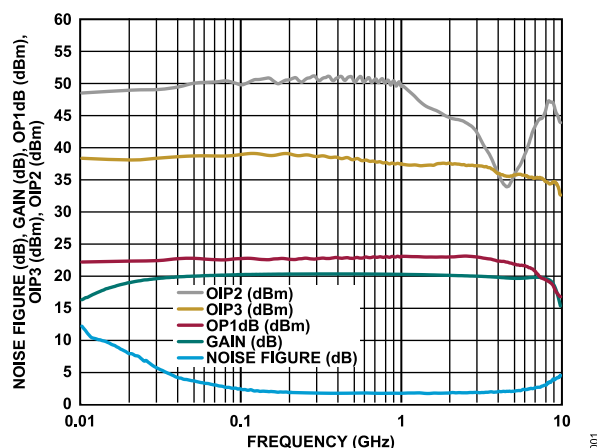


Figure 1. Frequency Response of HMC8413, 10MHz to 9GHz LNA

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REVISION HISTORY**9/2025—Revision 0: Initial Version**

DEPLETION MODE LNAS VS. SINGLE POSITIVE SUPPLY ENHANCEMENT MODE LNAS

One fundamental differentiator between LNA's from Analog Devices is how they are biased, specifically how the quiescent operating current is set. LNA's that are fabricated on depletion mode processes, have a drain to source (or V_{DD} to ground) resistance of almost 0Ω when the gate voltage is open or equal to $0V$ (see [Figure 2](#)). This low resistance between V_{DD} and ground can be physically measured using an ohmmeter and is often mistaken as an indicator that the device is damaged or destroyed.

A negative gate voltage must be applied to bias on the main transistor. This voltage is typically between $-0.5V$ and $-2V$ and must be carefully sequenced so that it is applied before V_{DD} turns on and removed after V_{DD} turns off.

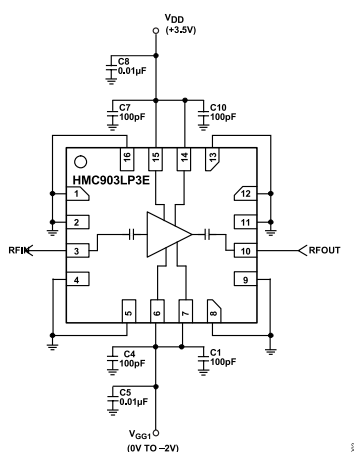


Figure 2. Depletion Mode LNA

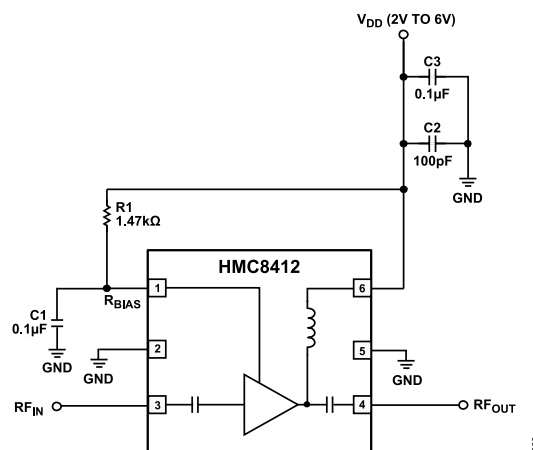


Figure 3. Enhancement Mode LNA

[Figure 3](#) shows the application circuit for a typical enhancement mode LNA. In this case, the operating bias current is set by connecting a resistor between the R_{BIAS} pin and V_{DD} . All newer LNA's from Analog Devices use this circuit architecture and have part numbers with the format HMC841X, ADL7XXX, and ADL8XXX (for example, [HMC8414](#), [ADL7078](#), and [ADL8100](#)). Power supply sequencing is not critical and because there is no need for a negative supply voltage, these LNA's are easier to use than depletion mode devices.

There are also a class of older LNA's that have a fixed bias setting. These devices typically have a single positive supply voltage and a fixed quiescent bias current that cannot be adjusted.

THE ANALOG DEVICES LNA PORTFOLIO

Analog Devices portfolio of LNAs is a combination of multiple product portfolios developed or acquired over time. The family of

approximately 200 devices is summarized in [Table 1](#) with the newer devices and product families at the top and the older devices at the bottom.

Table 1. Analog Devices RF LNA portfolio

Part Numbering	Frequency Coverage	Frequency Bands	Target End Applications	Example	Comments	Approximate Number of Generics
ADL81XX	10MHz to 54GHz	VHF to V	General purpose	ADL8100	Single supply with RBIAS	20
ADL7XX	1GHz to 20GHz, 50GHz to 95GHz	L to KU, V, E	General purpose	ADL7078		2
ADL900X	10 MHz to 28GHz	VHF to K	General purpose	ADL9005	Single supply	4
HMC841X	10MHz to 11GHz	HF to X	General purpose	HMC8411	Single supply with RBIAS	9
HMC840X	10MHz to 30GHz	HF to K	General purpose	HMC8400-DIE	Requires negative gate voltage	3
ADL8111X, HMC8414	10MHz to 25.5GHz	HF to K	High dynamic range receivers, bidirectional circuits, test and measurement equipment	ADL8113	LNA with bypass and bidirectional operation, single supply with RBIAS	4
ADLXXX-CSL, ADLXXX-CSH	30KHz to 31GHz	LF to K _A	Commercial space low, commercial space high	ADL8141-CSL	Single supply with RBIAS	5
ADHXXXS	10MHz to 65GHz	HF to V	Traditional space	ADH903S	QMLV space qualified	18
ADL57XX	6GHz to 24GHz	C, X, K _U , K	Differential Rx mixer driver, ADC driver	ADL5721	Single-ended to differential amp, single supply	5
MAX-XXXX	40MHz to 4GHz	VHF, UHF, L, S	GPS/GNSS, automotive FM radio, general purpose	MAX2694	Single supply silicon based devices	40
HMC-ALHXXX	500MHz to 86MHz	L to E	General purpose	HMC-ALH244-DIE	Older portfolio, die product only	15
HMC1XXX, HMC2XX, HMC3XX, HMC4XX, HMC5XX, HMC6XX, HMC7XX, HMC9XXX, ADL55XX	30KHz to 36GHz	LF to V	General purpose	HMC1126 , ADL5521	Older portfolio, mostly require negative gate voltage	57

Newer devices (for example those with part numbers ADL8XXX, ADL7XXX, and HMC8XXX) tend to be broadband, internally matched and operate from a single positive supply voltage. These have the numbering format of AD7XXX, ADL8XXX, ADL9XXX, HMC840X, and HMCH41X (for example, [ADL7078](#), [ADL8102](#), [ADL9005](#), [HMC8402](#), and [HMC8411](#)). An example is shown in [Figure 4](#).

Devices with the numbering format [HMC8414](#) and ADL811X (for example [ADL8112](#)) are LNAs with bypass and bidirectional operating modes, which are useful in high dynamic range receiver applications, bidirectional radar circuits, and in test and measurement equipment (see example in [Figure 5](#)).

The majority of these newer devices operate on a single positive supply voltage with a bias current that is set by an external resistor (R_{BIAS}).

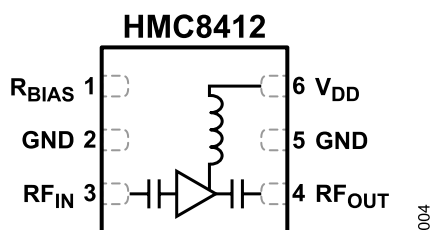


Figure 4. Typical Devices from Analog Devices LNA Portfolio, [HMC8412](#)

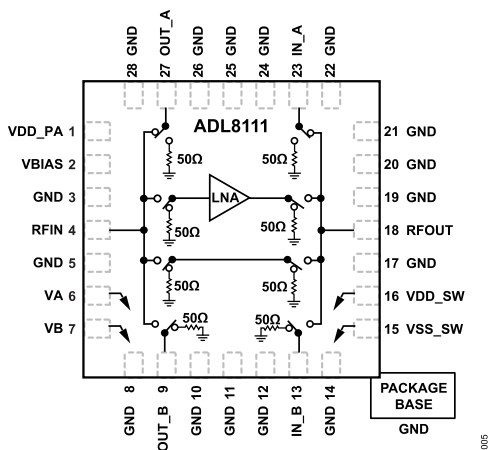


Figure 5. Typical Devices from Analog Devices LNA Portfolio, [ADL8111](#)

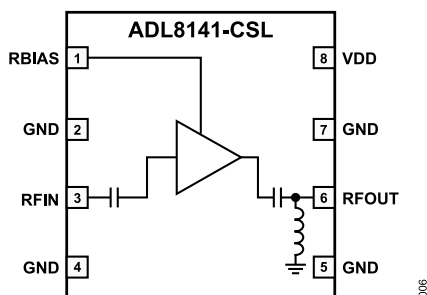


Figure 6. Typical Devices from Analog Devices LNA Portfolio, [ADL8141-CSL](#)

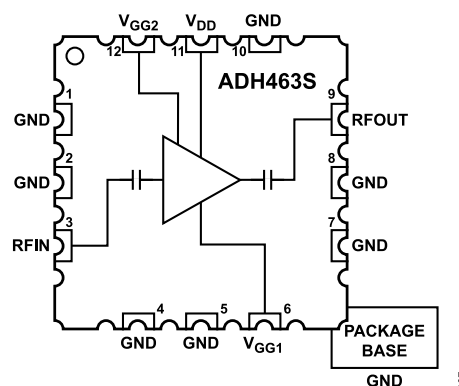


Figure 7. Typical Devices from Analog Devices LNA Portfolio, [ADH463S](#)

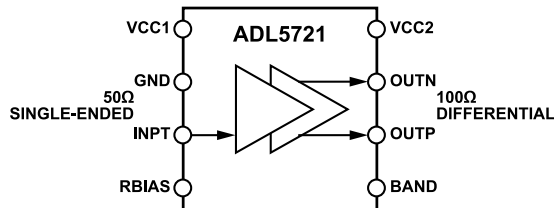


Figure 8. Typical Devices from Analog Devices LNA Portfolio, [ADL5721](#)

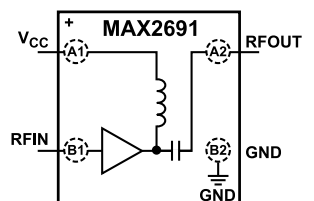


Figure 9. Typical Devices from Analog Devices LNA Portfolio, [MAX2691](#)

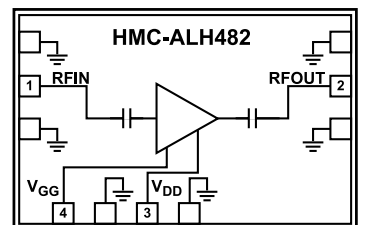


Figure 10. Typical Devices from Analog Devices LNA Portfolio, [HMC-ALH482](#)

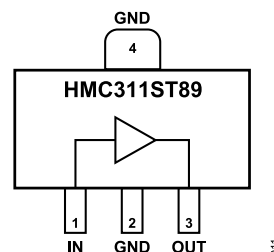


Figure 11. Typical Devices from Analog Devices LNA Portfolio, [HMC311ST89](#)

Analog Devices family of commercial space LNAs have the designation -CSL (commercial space low) or -CSH (commercial space high) at the end of their part numbers (see example in [Figure 6](#)). Traditional fully space qualified products from Analog Devices

feature part numbers which begin with ADH and end with the letter S (see example in [Figure 7](#)). For legacy Hittite devices that are available in commercial and space qualified versions, the part numbering is HMCXXX and ADHXXXS respectively (for example [HMC903LP3E](#) and [ADH903S](#) are the part numbers for the commercial and space qualified version of the HMC903 6GHz to 17GHz LNA).

The ADL57XX family are LNAs with single ended inputs and differential outputs (for example [ADL5721](#)). A typical device is shown in [Figure 8](#). These devices, which operate from 6GHz to 24GHz, are developed to drive RF mixers with differential inputs as well as RF sampling analog-to-digital converters (ADC) such as Analog Devices mixed signal front end (MxFE) family. These silicon based devices are all internally matched and operate from a single positive supply voltage.

MAX part numbers with four or five digits (for example [MAX2657](#) and [MAX12000](#)) are an older family of devices mostly developed for specific applications such as automotive FM radio, global positioning system (GPS), and global navigation satellite systems (GNSS) receivers. These silicon based devices are all internally matched and operate from a single positive supply voltage (see example in [Figure 9](#)).

The HMC-ALHXXX-DIE family (for example [HMC-ALH216-DIE](#)) are older devices, which are only offered in die form, that feature a frequency coverage from 500MHz to 86GHz (see example in [Figure 10](#)). These devices are internally matched and require a negative gate biasing voltage.

LNAs with three digit part numbers (for example [HMC311ST89](#)) along with the HMC1XXX and ADL55XX belong to a large older family of general purpose devices (see example in [Figure 11](#)). The majority but not all of these devices are internally matched. Moreover, the majority of these devices require a negative gate voltage with the remainder running from a single positive supply with a fixed bias.

Approximately 70 parts from Analog Devices family of 200 LNAs are in die form factor. These devices feature the same functionality as the packaged versions of the parts, but these parts are not in an integrated circuit (IC) package. LNAs in die form tend to have a slightly higher gain and bandwidth, and a lower noise figure than the equivalent packaged version. The nomenclature on LNAs in die form is fairly inconsistent. For example, [ADL8106CHIPS](#), [HMC1040-DIE](#), and [HMC1127](#) are all part numbers for LNAs in die form. To find and differentiate all suitable parts, use the Analog Devices parametric search tool and sort via the package option.

DEVICE SELECTION BASED ON AGE AND PRODUCTION STATUS

When selecting an LNA, it is worthwhile to check the age of the device and its production status. This information can be found on the device product page or in the parametric search table. The

launch date indicates when the device went into mass production. It is generally recommended to select the newest device that meets your technical needs. [Table 2](#) lists the various production status indicators and its meaning.

Table 2. Analog Devices Production Status Definitions

Production Status	Interpretation
Recommended for new designs	This product has been released to the market. For new designs, Analog Devices recommends the usage of these products.
Production	At least one model within this product family is in production and available for purchase. The product is appropriate for new designs but newer alternatives may exist.
Not recommend for new designs	This designates products that Analog Devices does not recommend broadly for new designs.
Last time buy	This device is going out of production and this is an opportunity to stock up if device is still currently utilized.
Obsolete	The models in this product family are no longer available. The product page for the device may suggest a substitute.

LNAs VS. GAIN BLOCKS VS. DRIVER AMPLIFIERS

The terms LNA, gain block, and driver amplifier are used widely in the industry but are not well defined. There is no agreed upon definition for what constitutes an LNA, that is, there is no threshold for the device noise figure to be below a certain level before it can be called an LNA.

Gain blocks are generally understood to be relatively low power RF amplifiers, which are internally matched and exhibit a relatively wide operating frequency range (for example at least five octaves).

However, a number of LNAs from Analog Devices meet this criteria. Driver amplifiers are generally used to drive ADCs or RF power amplifiers. In both cases, the output specifications such as third order intercept (OIP3), second order intercept (OIP2), and compression (OP1dB) are critical to the end application. However, based on Analog Devices LNA portfolio, a number of devices feature an excellent output referred specifications and can, therefore, be used as driver amplifiers. This suggests that when looking for a device, focus more on the target specifications rather than what the device is defined as. These specifications are discussed further in next sections.

CRITERIA FOR SELECTING A LOW NOISE AMPLIFIER

FREQUENCY RANGE

The specified frequency range for an LNA is the range over which Analog Devices has judged the device to work well. This specified frequency range is primarily determined by the behavior of gain, noise figure, and return loss of the device versus frequency. The degradation of these characteristics outside of the specified frequency range is generally graceful. Also, a number of data sheets and s-parameter data sets include data that goes beyond the specified frequency range. This allows the end user to judge and decide whether to use the device outside of the specified range. Figure 12 shows the gain and return loss versus frequency for the HMC8413, a 10kHz to 10GHz LNA. The plot is scaled to 12GHz so that the behavior above 10GHz is apparent.

Due to the internal broadband matching on most devices, it is generally not possible to extend the frequency range of devices through the use of external matching circuitry. For broadband devices that require an external drain biasing inductor (such as the HMC8413), the choice of an inductor can reduce the operating frequency range. The AN-2061: Application Note, *Wideband Bias Tee Design Using 0402, SMD Components* details more on this topic. It presents a surface-mount bias tee circuit that has a flat response beyond 20GHz for broadband amplifiers that require external bias inductors (a bias tee is a circuit used to provide DC bias current and AC coupling; it consists of a series capacitor and a shunt inductor that is usually connected to the positive power supply).

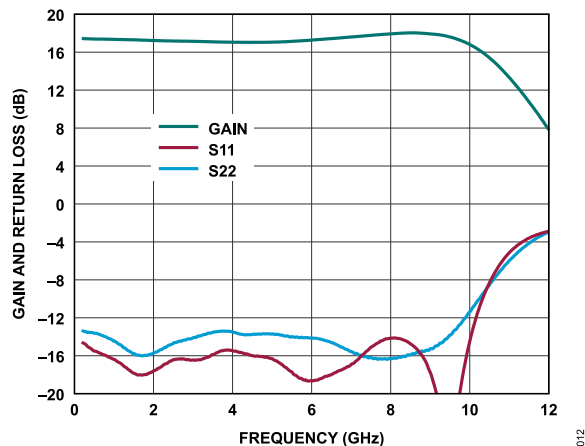


Figure 12. ADL8122 Gain and Return Loss vs. Frequency

GAIN

LNAs typically have a fixed gain. If a precise gain is required, the most practical way to achieve this is to choose an amplifier with a gain that is slightly higher than necessary and then add attenuation to the circuit. This can be achieved using a T attenuator (a resistor based attenuator that consists of a series resistor and a shunt resistor followed by another series resistor) or a Pi attenuator (a resistor based attenuator that consists of a shunt resistor and a series resistor followed by another shunt resistor) at the input and/or output of the amplifier or by including a digital step attenuator (DSA)

in the signal chain. If the LNA is at or close to the start of the signal chain where its noise figure has a large impact, any gain reducing attenuation must be placed after the LNA.

Adjusting the bias current has a slight impact on the gain, but should not, in general, be used as a gain control mechanism. It generally requires significant movement of the bias current to affect a relatively small change in gain. This bias current adjustment tends to have a larger impact on compression and distortion.

CURRENT CONSUMPTION, I_{DQ} AND I_{DD}

Recent Analog Devices LNA data sheets make a distinction between quiescent and dynamic current consumption. Quiescent current (I_{DQ}) is the current consumption when the amplifier is biased on but no RF signal is present. Dynamic current consumption (I_{DD}) is the current consumption when an RF signal is present. I_{DD} is usually but not always higher than I_{DQ} . When dimensioning power management systems, it is important to budget for enough current to support the highest expected RF output power. This includes active bias control circuits where the continuous bias current must be set to support maximum expected output power. Figure 13 shows current consumption, gain, output power, and power added efficiency (PAE) versus input power for the ADL8107, a 6GHz to 18GHz LNA. In this case, I_{DQ} is 90mA and I_{DD} maxes out at 123mA when the input and output power are at +5dBm and +20dBm respectively. Note, that the behavior of I_{DD} versus output power tends to change with signal frequency.

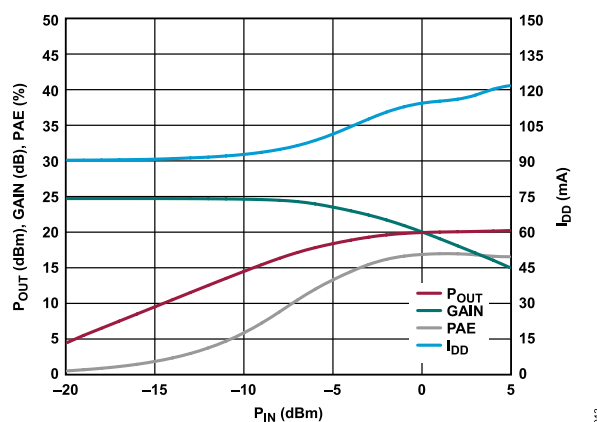


Figure 13. ADL8107 Power Sweep at 12GHz

NOISE FIGURE

After frequency range, the noise figure of an LNA is among the most critical specifications. A data sheet publishes a typical noise figure on its front page with specifications at various spot frequencies in the specification tables and plots of noise figure versus frequency and other variables in the typical performance characteristics section of the data sheet.

As frequency increases, noise figure tends to degrade steadily. As frequency decreases, noise figure tends to improve gradually before rising steeply at a certain break point that is usually in the tens or hundreds of MHz. Since most LNAs from Analog Devices

are internally matched, this performance cannot be improved using external matching circuitry.

For devices that have an adjustable supply voltage range (for example 3V or 5V), adjusting the supply voltage may impact the noise figure. Moreover, adjusting the bias current might or might not have an impact on noise figure. Figure 14 shows a typical plot of noise figure versus frequency and bias current. In this case, the noise figure of ADL8140, a 10GHz to 18GHz LNA, improves by about 0.3dB as the bias current is increased from 20mA to 45mA. Changing the supply voltage of this device from 1.2V to 3.5V while keeping the bias current at 35mA results in only a 0.1dB improvement in noise figure. However, there are other devices where adjustments on supply voltage or bias current have no noticeable impact on noise figure. Thus, it is best to consult the data sheet of the device you are considering to use to see how the device behaves.

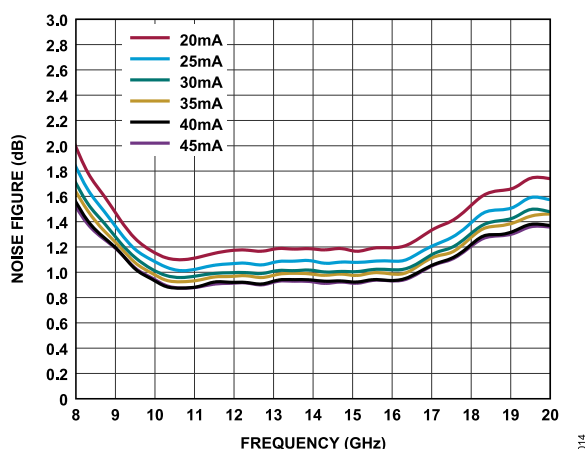


Figure 14. Noise Figure vs. Frequency and Bias Current for ADL8140, 10GHz to 18GHz LNA

RESIDUAL PHASE NOISE

Residual phase noise is a measure of how much phase noise is added by the amplifier. It is a critical specification for amplifiers that are being used as local oscillator buffers or in radar receivers because the amplifier can add phase noise to the system and, thereby, desensitize it. Residual phase noise is measured by observing the phase noise at the output of the amplifier and canceling out the phase noise of the input signal driving the amplifier. Residual phase noise tends to vary with the RF output power level of the amplifier. It often improves when the output power is at compression or saturation compared to when the amplifier is operating well backed from compression. However, this behavior is not universal.

Residual phase noise is typically presented in data sheets for newer devices in plots at multiple frequencies at P_{SAT} , OP1dB (output 1dB compression point in dBm), and with the output power backed off approximately 10dB from compression. An example plot is shown in Figure 15. In this case, the phase noise improves as the output power increases. However at saturation, the phase noise degrades slightly at some frequency offsets compared to the phase noise at compression. Note that the apparent phase noise increase above 1MHz offset frequency is most likely a measurement artifact

caused by imperfect cancellation of the source phase noise in the setup.

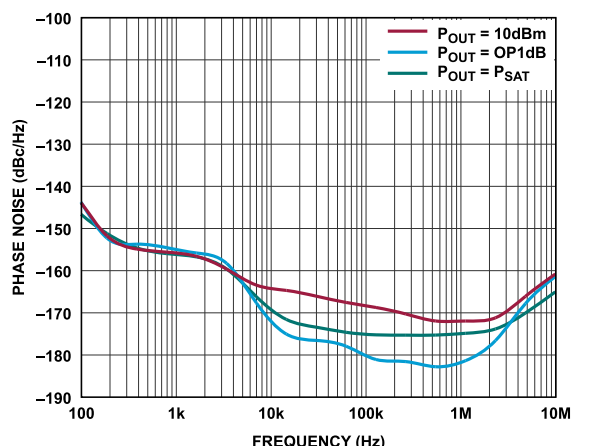


Figure 15. Residual Phase Noise of AD8154, a 10MHz to 6GHz Low Phase Noise LNA

SECOND AND THIRD ORDER INTERMODULATION

The second and third order intercept points, which are commonly referred to as OIP2 (output second order intercept point in dBm) and OIP3 (output third order intercept point in dBm), are indicators of the second order ($F_1 + F_2$, $F_1 - F_2$) and third order ($2F_1 - F_2$, $2F_2 - F_1$) intermodulation products that the device produces. For LNAs from Analog Devices, OIP2 and OIP3 are generally measured using a 1MHz separation between the two input carriers at a power level that is typically 10dB to 20dB below the 1dB compression point. This level is specified in the specification table, and some data sheets provide IP3 and IMD3 data (in dBm and dBc) at different power levels.

SECOND AND THIRD HARMONIC

In wideband applications or applications where the LNA is driving an ADC, the size of the second and third harmonic might be important. Some Analog Devices data sheets contain plots of second and third harmonic versus frequency, but most do not. Where harmonic data is not available, the second and third order intermodulation specifications can be used to estimate the magnitude of the second and third harmonics using the following equations:

$$HD2 \text{ (dBc)} = P_O - OIP2 - 6 \quad (1)$$

$$HD3 \text{ (dBc)} = 2 \times (P_O - OIP3) - 9.54 \quad (2)$$

Where:

P_O is the power of the fundamental in dBm.

OIP2 and OIP3 are the second and third order intercept points in dBm.

Note that HD2 refers to second harmonic in dBc while HD3 refers to third harmonic in dBc.

COMPRESSION AND SATURATION

As you increase the RF input power to an RF amplifier, the output power rises proportionally until it starts to compress. The 1dB compression point is the point when the RF output power level and gain are 1dB lower than they should be. Analog Devices generally specifies output referred 1dB compression or OP1dB. Output referred compression is related to input referred compression (IP1dB) by the following equation:

$$\text{OP1dB} = \text{Small Signal Gain} + \text{IP1dB} - 1 \quad (3)$$

The output saturation point (P_{SAT}) is reached when the RF output power no longer responds to increased input power. For GaAs and silicon devices, P_{SAT} is typically 1dB to 3dB higher than the compression point (for GaN devices the delta between OP1dB and P_{SAT} tends to be larger).

For LNAs whose power supply voltage can be varied (for example, if the LNA supports 3.3V and 5V operation), OP1dB and P_{SAT} is generally higher at higher supply voltages. For devices that have an adjustable bias current (either via a negative gate voltage or an R_{BIAS} resistor), OP1dB can be significantly impacted by the bias setting with higher bias currents generally resulting in a higher OP1dB. An example of this is shown in Figure 16.

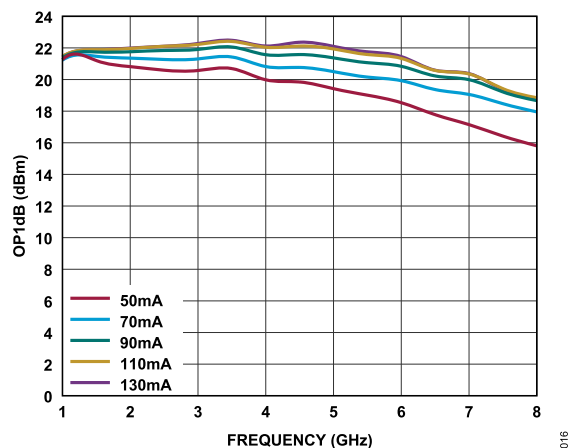


Figure 16. ADL8108, 1GHz to 8GHz LNA, OP1dB vs. Bias Current

While higher supply voltage tends to increase P_{SAT} , the relationship between P_{SAT} and bias current varies by device. In some cases, increasing the current also increases the P_{SAT} but the P_{SAT} levels of the other devices are less responsive to increased bias currents. An example of this shown in Figure 17.

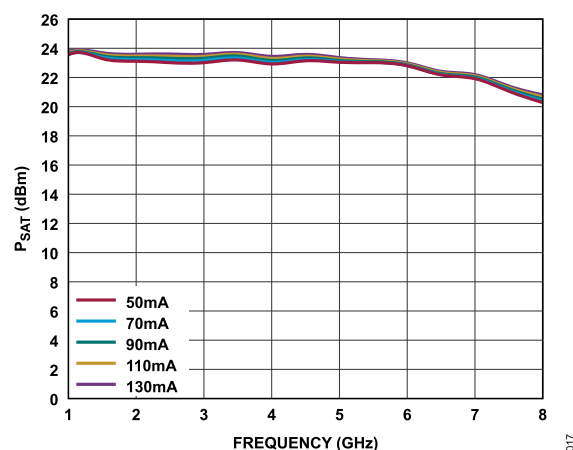


Figure 17. ADL8108, 1GHz to 8GHz LNA, P_{SAT} vs. Bias Current

OVERDRIVE RECOVERY

Overdrive recovery is a measure of how quickly an amplifier returns to its normal operation after it has been driven deep into saturation. In this case, the RF input power is high enough to drive the amplifier into saturation while not exceeding the absolute maximum rating specifications for an RF input power. Overdrive recovery is measured by alternating the input level between a high power signal (often called the blocking signal) that puts the amplifier into saturation followed by a lower level signal (usually at a different frequency) that brings the amplifier back to linear operation. The time it takes for the output to switch from deep saturation back to linear operation is then measured.

When the blocking signal is present, the measured input to output gain is significantly reduced. Once the blocking signal is removed, the input to output gain eventually returns to its specified small signal level. Overdrive recovery is usually defined as the time from removal of the blocker to the gain returning to 90% of its small signal value.

Overdrive recovery times tend to vary as a function of the level of the blocking signal. As a result, the data sheet usually provides a plot of overdrive recovery time versus blocking level instead of a specification in the specification table (see Figure 18 for an example). A data sheet typically includes one overdrive recovery plot, which is taken at one frequency. For wideband devices, the overdrive recovery behavior does not change markedly versus frequency, power supply level, or bias current setting.

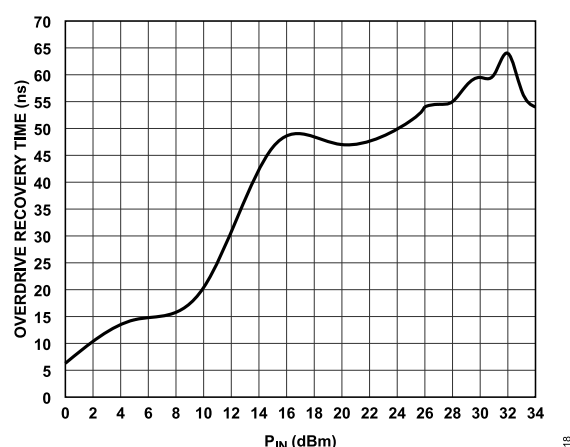


Figure 18. Typical Overdrive Recovery vs. Blocking Signal Level Plot (Taken from ADL7078 Data Sheet)

ENABLE/DISABLE RESPONSE TIME

In some power sensitive applications, it is necessary to disable the LNA when it is not in use. Some newer LNAs, such as the ADL8108, include a digital input pin, which can be used to power up and power down the device in 15ns and 10ns respectively. Where this function is not available, the RBIAS pin can be used as an enable/disable function. However, the enable/disable response can vary because of the device architecture. This is described in further detail in AN-2599: Application Note, Using the RBIAS Pin of Single Positive Supply RF Amplifiers as a Fast Enable/Disable Input.

INPUT SURVIVABILITY AND MAXIMUM RF INPUT POWER

While noise figure and input sensitivity are the most critical specifications of an RF receiver, input survivability is also important, particularly in applications such as radar and electronic warfare where deliberate jamming using high powered signals is common. The input survivability of an LNA is specified in dBm in the absolute maximum ratings section of the data sheet. LNAs that can survive relatively high input power levels are specifically referred to as devices with high input survivability in its product descriptors.

The input survivability of the LNA, when the RF input signal is pulsed, tends to be slightly higher (typically 1dB to 3dB). Survivability versus pulsed input power is generally not specified in the device data sheet. If it is specified, this information is provided in the absolute maximum ratings section.

ELECTROSTATIC DISCHARGE (ESD) RATING

Information about the ESD susceptibility of the LNA is found in the absolute maximum ratings section of the data sheet. Most data sheets only provide a human body model (HBM) specification. Table 3 summarizes the voltage ratings for the various HBM ESD ratings. If a device is rated Class 1A for example, this means that it survived a 250V HBM pulse and was damaged by a 500V pulse. If no ESD rating is included in the data sheet, assume that the rating is Class 0.

Table 3. HBM ESD Classification

HBM ESD Class	Voltage Range
Class 0	0V to <250V
Class 1A	250V to <500V
Class 1B	500V to <1,000V
Class 1C	1,000V to <2,000V
Class 2	2,000V to <4,000V
Class 3A	4,000V to <8,000V
Class 3B	≥ 8,000 volts

THERMAL RESISTANCE

Thermal resistance of an integrated circuit is a measure of how easily heat generated in the die dissipates into the outside world. In the case of most newer LNAs, the die sits on a metal slug, which is exposed on the underside of the device (see Figure 19). As a result, the primary heat flow path is from the die to the metal slug to the printed circuit board (PCB). This assumes that the metal slug of the device makes good thermal contact with a metal pad on the top layer of the PCB, which has an onward heat conduction path (for example, vias which go through the board and connect to a heat sink). In this scenario, there is very little heat conduction from die to air, that is, through the top of the package or from the die to the pins. As a result, most RF amplifier data sheets just specify junction to case thermal impedance.



Figure 19. Typical Lead Frame Chip Scale Package (LF CSP) Package

A high junction-to-case thermal resistance is not necessarily problematic. The ultimate goal of thermal resistance calculations is to establish that the die is not exceeding its specified maximum junction temperature (usually 175°C for GaAs amplifiers). The equation for calculating the junction temperature is as follows:

$$T_{\text{JUNCTION}} = T_{\text{CASE}} + \theta_{\text{JC}} \times P_{\text{DISS}} \quad (4)$$

For example, if θ_{JC} is 100°C/W and power dissipation is 100mW, the junction temperature is 95°C if the case is at its maximum temperature of 85°C. In this case, the device has a high thermal resistance but the junction temperature is at a safe level because the power consumption is relatively low. This topic is discussed further in AN-2591: Application Note, When It Comes to Long-Term Reliability of RF Amplifier ICs, Focus First on Die Junction Temperature.

As most heat flow is through the bottom of the package and into the PCB, any heat sinking is generally implemented on the underside of the PCB. A heat sink can be added to the top side of the package, but it is generally of little benefit because the junction-to-case top thermal resistance is much higher than the junction-to-case bottom thermal resistance (θ_{JC}).

POWER SUPPLY REQUIREMENTS AND SEQUENCING

LNAs developed on depletion mode and enhancement mode process differ in the key respect that the depletion mode devices require a negative gate voltage and the enhancement mode devices do not. As this has a significant impact on the power supply design, it must be taken into consideration when choosing a device.

Figure 20 shows a typical power management circuit for an enhancement mode LNA. In this example, the [LT8607](#) step-down regulator is used to step down a 12V rail to 5.5V, which is then applied to the [LT3042](#) low dropout (LDO) linear regulator, which generates a low noise 5V output and suppresses the switching spurs from the buck regulator. The LT8607 can accept input voltages as high as 42V and can source up to 750mA. The LT3042 low dropout regulator can source up to 200mA. If power supply voltage is being developed as a bus supply to serve multiple components, higher current devices can be used. The [LT8608](#) and [LT8609](#) step-down regulators can source maximum currents of 1.5A and 3A respectively. These devices are pin compatible with the LT8607. The [LT3045](#) linear regulator, which is pin compatible with LT3042, can source up to 500mA.

Figure 21 shows an application circuit for the [ADL8106](#), an 18GHz to 54GHz LNA. In this circuit, the [HMC920](#) active bias controller

is being used to generate the positive supply voltage (VDD) and the negative gate voltage (VGG). The active bias controller safely sequences these two voltages and uses negative feedback to adjust the gate voltage until the desired set point drain current is flowing. This current is set by a resistor connected to the HMC920 (R4 in the schematic), and must be set high enough to provide the highest expected drain current for the application, which always flows (140mA in this case). HMC920 can operate from supply voltages from 5V to 16.5V. It can generate drain voltages from 3V to 15V. The available gate voltage range is 0V to -2.5V, and the drain current can be set between 0mA and 500mA.

As an alternative to an active bias control circuit, a fixed negative voltage generator can also be used, such as the [ADP5600](#) along with a positive LDO. However, for a number of reasons, this is not recommended. Fixing the gate voltage means that you are subject to part variations in the drain current. The relationship between drain current and specifications, such as compression and distortion, is tighter than their relationship to the gate voltage. If the same gate voltage is applied to multiple devices, each one has a slightly different drain current. More of this is discussed in [AN-1363: Application Note, Meeting Biasing Requirements of Externally Biased RF/Microwave Amplifiers with Active Bias Controllers](#). Thus, it is better for multiple devices to have the same drain current versus the same gate voltage. Moreover, you need to be careful with the sequencing. Make sure to use the power good (PG) output from the negative gate voltage generator to gate and enable the positive supply. This ensures that the gate voltage is present before the drain voltage.

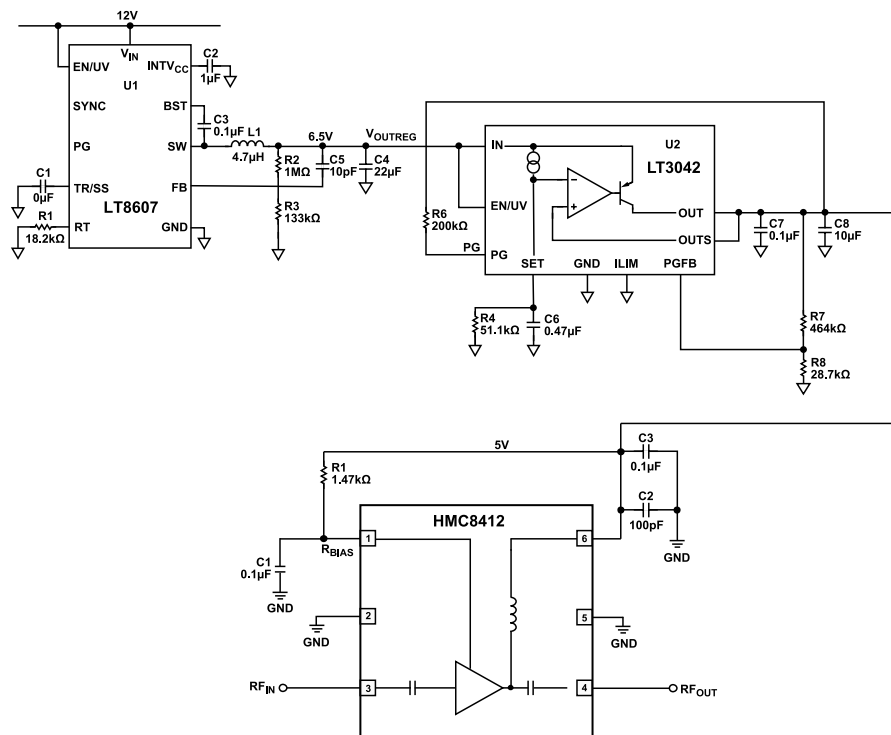


Figure 20. Generic Power Management Circuit for Single Positive Supply LNA with Resistor Programmable Bias Control

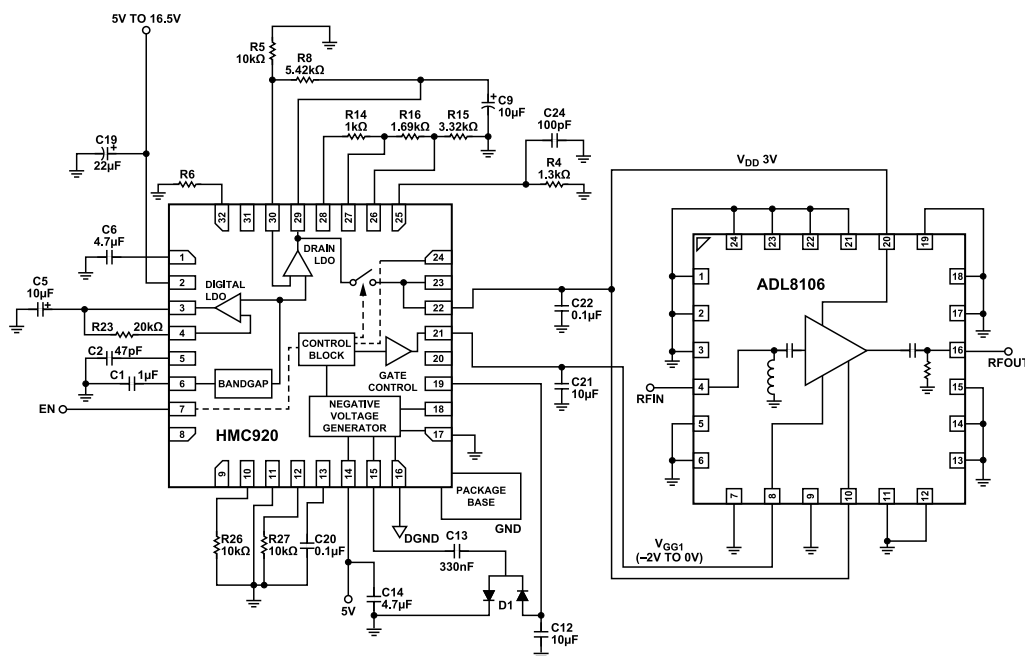


Figure 21. ADL8106 Wideband LNA with Negative Gate Voltage and Active Bias Control Using HMC920

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CHECKLIST FOR SELECTING AN LNA

Use the following checklist as a step by step guide to define your requirements and narrow down your options so that you select the device that best meets your needs.

1. Determine the primary specifications of the end application by the following parameters:
 - ▶ Gain
 - ▶ Frequency range
 - ▶ Noise figure
 - ▶ OIP3 and OIP2
 - ▶ Supply voltage and current consumption
2. Determine if the application requires a die or a packaged part.
3. [Use the product selection tool on Analog Devices website to narrow down the choices to four or five devices.](#) Try to choose devices that are listed as production or recommended for new designs. Avoid selecting not recommended for new designs or last time buy.
4. Determine the value of secondary specifications from the device data sheets:

REFERENCES

Soc, Ivan. Nash, Eamon. [AN-2061: Wideband Bias Tee Design Using 0402, SMD Components](#). Analog Devices, 2024.

Soc, Ivan. Nash, Eamon. [AN-2591: When It Comes to Long-Term Reliability of RF Amplifier ICs, Focus First on Die Junction Temperature](#). Analog Devices, 2024.

Bedrosian, Jim. Nash, Eamon. [AN-2560: Creating Amplifier Models in Keysight ADS Using Genesys and SystemVue Sys-Parameter Models](#). Analog Devices, 2023.

- ▶ RF input power survivability
- ▶ Overdrive recovery time
- ▶ Residual phase noise
- ▶ ESD rating

5. If you plan to perform simulations, download the s-parameters from the device product page. For Keysight Genesys and SystemVue users, the most recent sys-parameter library can be downloaded at www.analog.com/sys-parameters.
6. Finalize your device selection.

CONCLUSIONS

Analog Devices has an extensive portfolio of LNAs. To narrow down your options, focus on the latest devices that operate from a single positive supply (with production status that is equal to recommended for new designs or tagged as in production). These devices are simple to operate and only require a few external passive components to operate. If you choose an LNA that requires a negative gate voltage, biasing and safe sequencing can be provided by the [HMC920](#).

Kaya, Kagan. [AN-1363: Meeting Biasing Requirements of Externally Biased RF/Microwave Amplifiers with Active Bias Controllers](#). Analog Devices, 2022.

Smith, Dorant. [AN-2599: Using the RBIAS Pin of Single Positive Supply RF Amplifiers as a Fast Enable/Disable Input](#). Analog Devices, 2024.

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